Initial value problems with nonhomogeneous equations

We've already seen how to solve second order nonhomogeneous equations using the methods of undetermined coefficients and variation of parameters.

All we want to say here is that, once we've found the general solution as the sum of the complementary and particular solutions, we can still extend the problem one step further by solving an initial value problem.

Solving the IVP

To solve an initial value problem for a second order nonhomogeneous differential equation, we'll use undetermined coefficients or variation of parameters to find the general solution $y(x) = y_c(x) + y_p(x)$.

Then we'll differentiate the general solution to get $y'(x) = y_c'(x) + y_p'(x)$. Once we have the general solution and its derivative, we'll plug the initial conditions $y(x_0) = y_0$ and $y'(x_1) = y_1$ into these equations, which will create a system of linear equations that we can solve for c_1 and c_2.

Once we find c_1 and c_2, we'll substitute them back into the general solution, which will solve the initial value problem. Let's do an example, skipping straight to the point where we've already found the general solution.

Example

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Solve the initial value problem for the second order nonhomogeneous equation, given $y(0) = 0$ and $y'(0) = 1$.

$$y'' + y' = x - 2e^{-x}$$

If we solve the associated characteristic equation, we find roots $r = 0, \ -1$, which gives us the complementary solution with distinct real roots, $y_c(x) = c_1 + c_2 e^{-x}$.

Our guess for the particular solution will be $y_p(x) = Ax^2 + Bx + Cxe^{-x}$, and undetermined coefficients lets us find $A = 1/2$, $B = -1$, and $C = 2$, giving

$$y_p(x) = \frac{1}{2}x^2 - x + 2xe^{-x}$$

for the particular solution, and therefore a general solution of

$$y(x) = c_1 + c_2 e^{-x} + \frac{1}{2}x^2 - x + 2xe^{-x}$$

Differentiating the general solution gives

$$y'(x) = -c_2 e^{-x} + x - 1 + 2e^{-x} - 2xe^{-x}$$

Substitute the initial conditions $y(0) = 0$ and $y'(0) = 1$ into the general solution and its derivative.

$$0 = c_1 + c_2 e^{-(0)} + \frac{1}{2}(0)^2 - 0 + 2(0)e^{-(0)}$$

$$1 = -c_2 e^{-(0)} + 0 - 1 + 2e^{-(0)} - 2(0)e^{-(0)}$$

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Simplifying these equations gives

$$0 = c_1 + c_2$$

$$0 = c_2$$

Substitute $c_2 = 0$ into $0 = c_1 + c_2$.

$$0 = c_1 + 0$$

$$0 = c_1$$

With $c_1 = 0$ and $c_2 = 0$, the general solution becomes

$$y(x) = 0 + (0)e^{-x} + \frac{1}{2}x^2 - x + 2xe^{-x}$$

$$y(x) = \frac{1}{2}x^2 - x + 2xe^{-x}$$

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